
Session 3 (2h) — Assembly & Joints

Core concepts (1h)

- **Component vs body** — a body is raw geometry; a component is a body wrapped so it can move independently and be reused across designs. Assemblies are built from components.
- **Joints** (Fusion's assembly-constraint tool) — define how two components move relative to each other:
 - Rigid — fixed together, no motion (e.g., a lid screwed to a housing)
 - Revolute — rotates around one axis (e.g., a hinge)
 - Slider — moves along one axis (e.g., a drawer)
- **Ground component** — one component must be fixed in space as the reference everything else joints against — normally your main housing/base part.
- **Interference check** — Fusion can detect if two components physically overlap; run this before assuming a design will actually fit together.

On-site task link: 조립 (assembly)

This is the direct rehearsal for the on-site “조립” task — designing parts that snap, screw, or slide together correctly, not just standalone shapes.

Hands-on practice (1h)

1. Bring the Session 2 enclosure body into a new **assembly** design as a component; ground it.
2. Model a simple lid (rectangle matching the enclosure's opening, extruded 2mm thick) as a second component.
3. Add a **Rigid** joint positioning the lid on top of the enclosure.
4. Duplicate the lid, and this time add a **Slider** joint instead — observe how it can now slide in/out along one axis instead of being locked.
5. Run an interference check between the lid and the enclosure walls; if it flags overlap, adjust a dimension to fix it.

Checkpoint

- ☐ Can explain the difference between a body and a component
- ☐ Understands Rigid vs Revolute vs Slider joints and can name a real-world example of each
- ☐ Assembled at least two components with a joint and ran an interference check